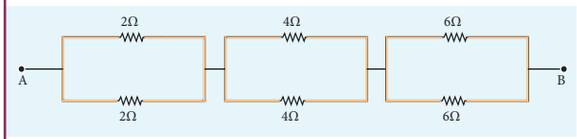
EXAMPLE 2.11

Calculate the equivalent resistance between A and B in the given circuit.



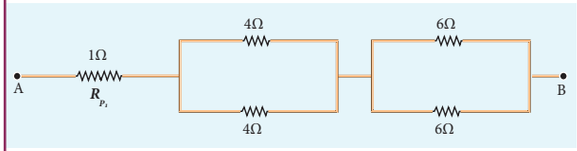
Solution

In all the sections, the resistors are connected in parallel.

Section 1

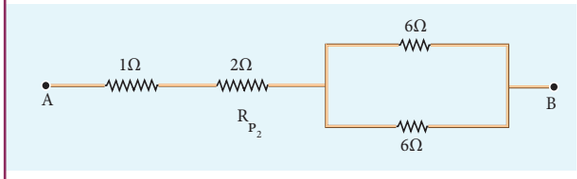
$$\frac{1}{R_{p_1}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{p_1}} = \frac{1}{2} + \frac{1}{2} = \frac{2}{2} \qquad R_{p_1} = 1\Omega$$



Section II

$$\frac{1}{R_{p_2}} = \frac{1}{4} + \frac{1}{4} = \frac{2}{4}, \quad \frac{1}{R_{p_2}} = \frac{1}{2}, \quad R_{p_2} = 2\Omega$$



Section III

$$\frac{1}{R_{p_3}} = \frac{1}{6} + \frac{1}{6} = \frac{2}{6}$$

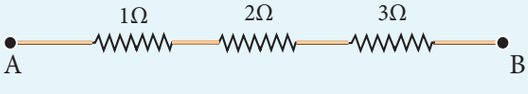
$$\frac{1}{R_{p_3}} = \frac{1}{3}, \quad R_{p_3} = 3\ \Omega$$

Equivalent resistance is given by

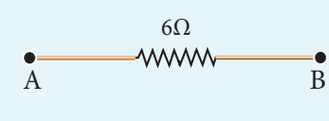
$$R = R_{p_1} + R_{p_2} + R_{p_3}$$

$$R = 1\ \Omega + 2\ \Omega + 3\ \Omega = 6\ \Omega$$

The circuit becomes,

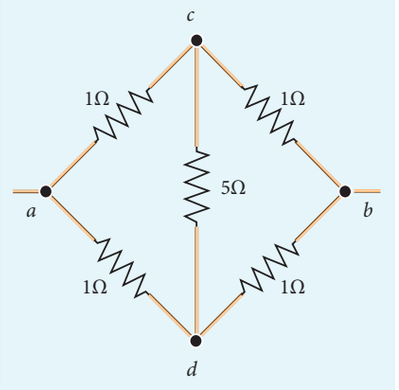


Equivalent resistance between A and B is



EXAMPLE 2.12

Five resistors are connected in the configuration as shown in the figure. Calculate the equivalent resistance between the points a and b.



Solution

Case (a)

To find the equivalent resistance between the points a and b, we assume that a current is entering the junction at a. Since all the resistances in the outside loop are the same (1Ω), the current in the branches ac and ad must be equal. Hence the points C and D are at the same potential and no current through 5 Ω. It implies that the 5 Ω has no role in determining the equivalent resistance and it can be removed. So the circuit is simplified as shown in the figure.

UNIT-2(XII-Physics_Vol-1).indd 93 14-02-2020 19:37:01